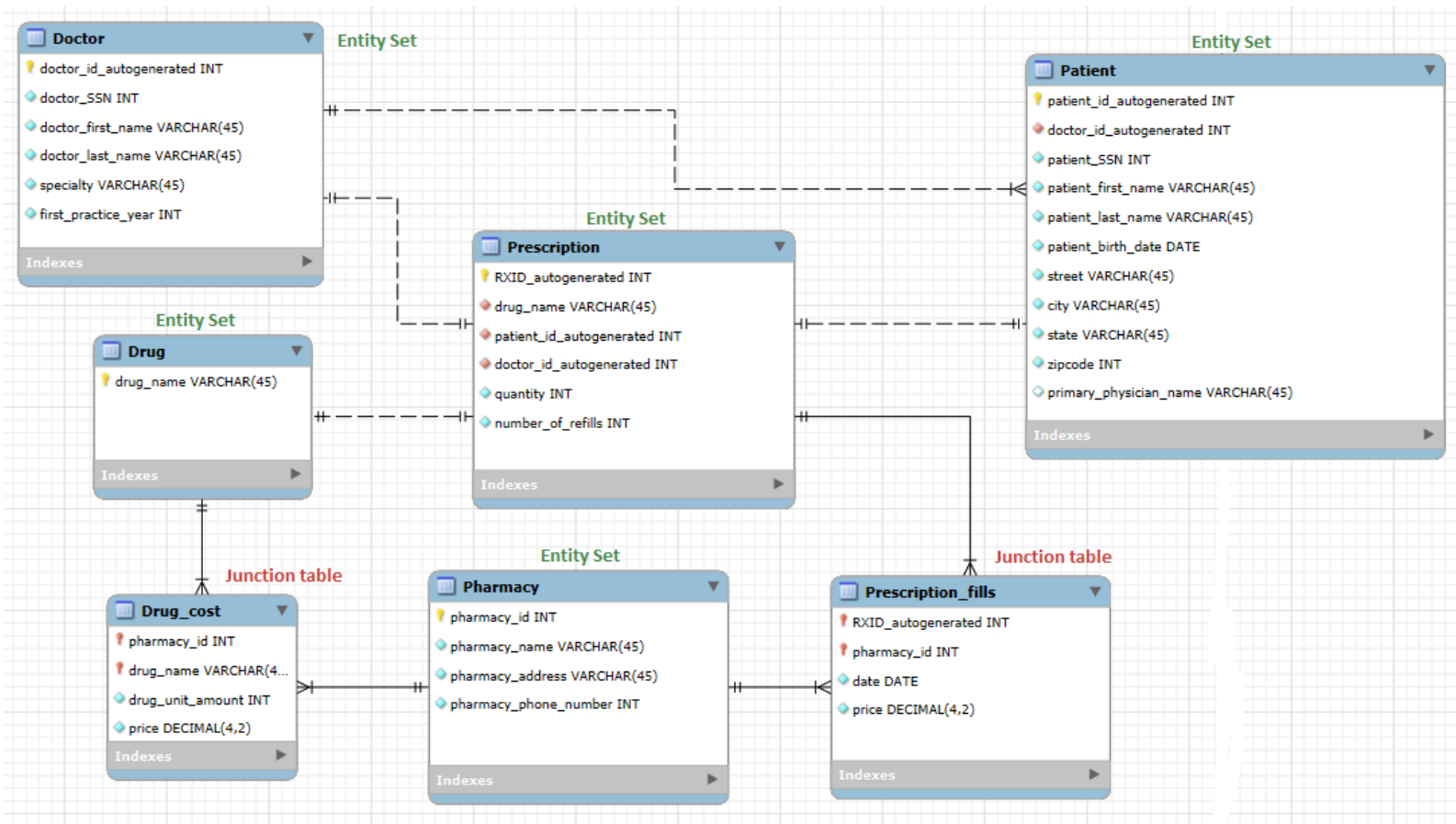


A Web Application to be used by Pharmacies

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ER Diagram (For this lab, we collectively agreed to use Wootark Kim's):



Schema (Created from MySQL Workbench):

```
-- MySQL dump 10.13  Distrib 8.0.42, for Win64 (x86_64)
--
-- Host: localhost    Database: lab19
-- -----
-- Server version 8.0.42

-- Create a database called lab19
CREATE DATABASE IF NOT EXISTS `lab19` /*!40100 DEFAULT CHARACTER SET
utf8mb4 COLLATE utf8mb4_0900_ai_ci */ /*!80016 DEFAULT ENCRYPTION='N' */;
USE `lab19`;

/*!40101 SET @OLD_CHARACTER_SET_CLIENT=@@CHARACTER_SET_CLIENT */;
/*!40101 SET @OLD_CHARACTER_SET_RESULTS=@@CHARACTER_SET_RESULTS */;
/*!40101 SET @OLD_COLLATION_CONNECTION=@@COLLATION_CONNECTION */;
/*!50503 SET NAMES utf8 */;
/*!40103 SET @OLD_TIME_ZONE=@@TIME_ZONE */;
/*!40103 SET TIME_ZONE='+00:00' */;
```

```

/*!40014 SET @OLD_UNIQUE_CHECKS=@@UNIQUE_CHECKS, UNIQUE_CHECKS=0 */;
/*!40014 SET @OLD_FOREIGN_KEY_CHECKS=@@FOREIGN_KEY_CHECKS,
FOREIGN_KEY_CHECKS=0 */;
/*!40101 SET @OLD_SQL_MODE=@@SQL_MODE, SQL_MODE='NO_AUTO_VALUE_ON_ZERO'
*/;
/*!40111 SET @OLD_SQL_NOTES=@@SQL_NOTES, SQL_NOTES=0 */;

--
-- Table structure for table `doctor`
--

DROP TABLE IF EXISTS `doctor`;
/*!40101 SET @saved_cs_client      = @@character_set_client */;
/*!50503 SET character_set_client = utf8mb4 */;
CREATE TABLE `doctor` (
  `id` int NOT NULL AUTO_INCREMENT,
  `ssn` varchar(9) NOT NULL,
  `last_name` varchar(30) NOT NULL,
  `first_name` varchar(30) NOT NULL,
  `specialty` varchar(30) DEFAULT NULL,
  `practice_since` int DEFAULT NULL,
  PRIMARY KEY (`id`),
  UNIQUE KEY `ssn` (`ssn`)
) ENGINE=InnoDB AUTO_INCREMENT=3 DEFAULT CHARSET=utf8mb3;
/*!40101 SET character_set_client = @saved_cs_client */;

--
-- Dumping data for table `doctor`
--

LOCK TABLES `doctor` WRITE;
/*!40000 ALTER TABLE `doctor` DISABLE KEYS */;
INSERT INTO `doctor` VALUES
(1,'111-1111','Dirt','Joe','Mullets',2001),(2,'222-2222','Gilmore','Happy'
,'Golf',1996);
/*!40000 ALTER TABLE `doctor` ENABLE KEYS */;
UNLOCK TABLES;

--
-- Table structure for table `drug`
--

```

```

DROP TABLE IF EXISTS `drug`;
/*!40101 SET @saved_cs_client      = @@character_set_client */;
/*!50503 SET character_set_client = utf8mb4 */;
CREATE TABLE `drug` (
  `drug_name` varchar(45) NOT NULL,
  PRIMARY KEY (`drug_name`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb3;
/*!40101 SET character_set_client = @saved_cs_client */;

--
-- Dumping data for table `drug`
--

LOCK TABLES `drug` WRITE;
/*!40000 ALTER TABLE `drug` DISABLE KEYS */;
INSERT INTO `drug` VALUES ('lisinopril');
/*!40000 ALTER TABLE `drug` ENABLE KEYS */;
UNLOCK TABLES;

--
-- Table structure for table `drug_cost`
--

DROP TABLE IF EXISTS `drug_cost`;
/*!40101 SET @saved_cs_client      = @@character_set_client */;
/*!50503 SET character_set_client = utf8mb4 */;
CREATE TABLE `drug_cost` (
  `pharmacy_id` int NOT NULL,
  `drug_name` varchar(45) NOT NULL,
  `drug_unit_amount` int NOT NULL,
  `price` decimal(4,2) NOT NULL,
  PRIMARY KEY (`pharmacy_id`,`drug_name`),
  KEY `fk_pharmacy_has_Drug_Drug1_idx` (`drug_name`),
  KEY `fk_pharmacy_has_Drug_pharmacy1_idx` (`pharmacy_id`),
  CONSTRAINT `fk_pharmacy_has_Drug_Drug1` FOREIGN KEY (`drug_name`)
REFERENCES `drug` (`drug_name`),
  CONSTRAINT `fk_pharmacy_has_Drug_pharmacy1` FOREIGN KEY (`pharmacy_id`)
REFERENCES `pharmacy` (`pharmacy_id`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb3;
/*!40101 SET character_set_client = @saved_cs_client */;

--

```

```

-- Dumping data for table `drug_cost`
--

LOCK TABLES `drug_cost` WRITE;
/*!40000 ALTER TABLE `drug_cost` DISABLE KEYS */;
INSERT INTO `drug_cost` VALUES (1,'lisinopril',999999,0.01);
/*!40000 ALTER TABLE `drug_cost` ENABLE KEYS */;
UNLOCK TABLES;

--
-- Table structure for table `patient`
--

DROP TABLE IF EXISTS `patient`;
/*!40101 SET @saved_cs_client      = @@character_set_client */;
/*!50503 SET character_set_client = utf8mb4 */;
CREATE TABLE `patient` (
  `patient_id_autogenerated` int AUTO_INCREMENT,
  `doctor_id_autogenerated` int,
  `patient_SSN` int NOT NULL,
  `patient_first_name` varchar(45) NOT NULL,
  `patient_last_name` varchar(45) NOT NULL,
  `patient_birth_date` date NOT NULL,
  `street` varchar(45) NOT NULL,
  `city` varchar(45) NOT NULL,
  `state` varchar(45) NOT NULL,
  `zipcode` int NOT NULL,
  `primary_physician_name` varchar(45) DEFAULT NULL,
  PRIMARY KEY (`patient_id_autogenerated`),
  KEY `fk_Patient_Doctor_idx` (`doctor_id_autogenerated`),
  CONSTRAINT `fk_Patient_Doctor` FOREIGN KEY (`doctor_id_autogenerated`)
REFERENCES `doctor` (`id`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb3;
/*!40101 SET character_set_client = @saved_cs_client */;

--
-- Dumping data for table `patient`
--

LOCK TABLES `patient` WRITE;
/*!40000 ALTER TABLE `patient` DISABLE KEYS */;

```

```

INSERT INTO `patient` VALUES
(1,1,1111111,'Jacobim','Mugatu','2001-01-01','3896 Meadow View
Drive','Manchester','CT',6040,'Joe Dirt');
/*!40000 ALTER TABLE `patient` ENABLE KEYS */;
UNLOCK TABLES;

--
-- Table structure for table `pharmacy`
--

DROP TABLE IF EXISTS `pharmacy`;
/*!40101 SET @saved_cs_client      = @@character_set_client */;
/*!50503 SET character_set_client = utf8mb4 */;
CREATE TABLE `pharmacy` (
  `pharmacy_id` int NOT NULL,
  `pharmacy_name` varchar(45) NOT NULL,
  `pharmacy_address` varchar(45) NOT NULL,
  `pharmacy_phone_number` int NOT NULL,
  PRIMARY KEY (`pharmacy_id`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb3;
/*!40101 SET character_set_client = @saved_cs_client */;

--
-- Dumping data for table `pharmacy`
--

LOCK TABLES `pharmacy` WRITE;
/*!40000 ALTER TABLE `pharmacy` DISABLE KEYS */;
INSERT INTO `pharmacy` VALUES (1,'CVS','1030 2nd Ave, Fairbanks, AK 99701,
US',1111111111);
/*!40000 ALTER TABLE `pharmacy` ENABLE KEYS */;
UNLOCK TABLES;

--
-- Table structure for table `prescription`
--

DROP TABLE IF EXISTS `prescription`;
/*!40101 SET @saved_cs_client      = @@character_set_client */;
/*!50503 SET character_set_client = utf8mb4 */;
CREATE TABLE `prescription` (
  `RXID_autogenerated` int NOT NULL AUTO_INCREMENT,

```

```

`drug_name` varchar(45) NOT NULL,
`patient_id_autogenerated` int NOT NULL,
`doctor_id_autogenerated` int NOT NULL,
`quantity` int NOT NULL,
`number_of_refills` int NOT NULL,
`create_date` date NOT NULL,
PRIMARY KEY (`RXID_autogenerated`),
KEY `fk_prescription_Drug1_idx` (`drug_name`),
KEY `fk_prescription_Patient1_idx` (`patient_id_autogenerated`),
KEY `fk_prescription_Doctor1_idx` (`doctor_id_autogenerated`),
CONSTRAINT `fk_prescription_Doctor1` FOREIGN KEY
(`doctor_id_autogenerated`) REFERENCES `doctor` (`id`),
CONSTRAINT `fk_prescription_Drug1` FOREIGN KEY (`drug_name`) REFERENCES
`drug` (`drug_name`),
CONSTRAINT `fk_prescription_Patient1` FOREIGN KEY
(`patient_id_autogenerated`) REFERENCES `patient`
(`patient_id_autogenerated`)
) ENGINE=InnoDB AUTO_INCREMENT=2 DEFAULT CHARSET=utf8mb3;
/*!40101 SET character_set_client = @saved_cs_client */;

```

```

--
-- Dumping data for table `prescription`
--

```

```

LOCK TABLES `prescription` WRITE;
/*!40000 ALTER TABLE `prescription` DISABLE KEYS */;
INSERT INTO `prescription` VALUES (1,'Crazy Pills',1,1,99999,99999,
CURDATE());
/*!40000 ALTER TABLE `prescription` ENABLE KEYS */;
UNLOCK TABLES;

```

```

--
-- Table structure for table `prescription_fills`
--

```

```

DROP TABLE IF EXISTS `prescription_fills`;
/*!40101 SET @saved_cs_client      = @@character_set_client */;
/*!50503 SET character_set_client = utf8mb4 */;
CREATE TABLE `prescription_fills` (
  `RXID_autogenerated` int NOT NULL,
  `pharmacy_id` int NOT NULL,
  `date` date NOT NULL,

```

```

    `price` decimal(4,2) NOT NULL,
    PRIMARY KEY (`RXID_autogenerated`,`pharmacy_id`),
    KEY `fk_prescription_has_pharmacy_pharmacy1_idx` (`pharmacy_id`),
    KEY `fk_prescription_has_pharmacy_prescription1_idx`
(`RXID_autogenerated`),
    CONSTRAINT `fk_prescription_has_pharmacy_pharmacy1` FOREIGN KEY
(`pharmacy_id`) REFERENCES `pharmacy` (`pharmacy_id`),
    CONSTRAINT `fk_prescription_has_pharmacy_prescription1` FOREIGN KEY
(`RXID_autogenerated`) REFERENCES `prescription` (`RXID_autogenerated`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb3;
/*!40101 SET character_set_client = @saved_cs_client */;

--
-- Dumping data for table `prescription_fills`
--

LOCK TABLES `prescription_fills` WRITE;
/*!40000 ALTER TABLE `prescription_fills` DISABLE KEYS */;
INSERT INTO `prescription_fills` VALUES (1,1,'2025-06-08',0.01);
/*!40000 ALTER TABLE `prescription_fills` ENABLE KEYS */;
UNLOCK TABLES;
/*!40103 SET TIME_ZONE=@OLD_TIME_ZONE */;

/*!40101 SET SQL_MODE=@OLD_SQL_MODE */;
/*!40014 SET FOREIGN_KEY_CHECKS=@OLD_FOREIGN_KEY_CHECKS */;
/*!40014 SET UNIQUE_CHECKS=@OLD_UNIQUE_CHECKS */;
/*!40101 SET CHARACTER_SET_CLIENT=@OLD_CHARACTER_SET_CLIENT */;
/*!40101 SET CHARACTER_SET_RESULTS=@OLD_CHARACTER_SET_RESULTS */;
/*!40101 SET COLLATION_CONNECTION=@OLD_COLLATION_CONNECTION */;
/*!40111 SET SQL_NOTES=@OLD_SQL_NOTES */;

-- Dump completed on 2025-06-08 10:40:12

```


Screenshots:

1. Register as a new patient with last name "Simpson", city "Rockville", zip code 62701 and a doctor with name "Spock". Show a successful registration.

Registration successful

Patient ID: 0
First Name: Homer
Last Name: Simpson
Birthdate: 1956-05-12
Street: 438 Test Street
City: Rockville
State: NT
Zipcode: 62701
Primary Physican: Spock

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2. Attempt to register as a new patient with last name "Burns" but with a doctor name that does not exist. Show a screenshot of the patient register form with the error message.

Primary Doctor not found

Your SSN:	123456780
Your First Name:	Tara
Your Last Name:	Burns
Birth Date:	01/01/2005 
Street:	123 Test Street
City:	Test
State:	TX
Zipcode:	99999
Primary Physician	
Name:	NameThatDoesNotExist
Register	

3. Create a prescription for the patient "Simpson" and doctor "Spock" for a drug "lisinopril" and quantity 90. Show the screen with the success message and prescription display.

Prescription created.

Rx:	3
Doctor ID:	8
First Name:	Spock
Last Name:	E
Patient ID:	2
First Name:	Homer
Last Name:	Simpson
Drug:	lisinopril
Quantity:	90
Refills remaining:	0
Pharmacy ID:	0
Name:	
Address:	
Phone:	
Date Filled:	
Cost:	\$

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4. Attempt to create a prescription with an invalid drug name. Show a screen with the create prescription form and error message.

New Prescription Form

Drug not found

Doctor ID:	1
Doctor First Name:	Joe
Doctor Last Name:	Dirt
Patient ID:	1
Patient First Name:	Jacobim
Patient Last Name:	Mugatu
Drug Name:	advil
Quantity:	2
Number of refills:	2

Create Prescription

New Prescription Form

Drug not found

Doctor ID:

Doctor First
Name:

Doctor Last
Name:

Patient ID:

Patient First
Name:

Patient Last
Name:

Drug Name:

Quantity:

Number of refills:

5. Attempt to fill a prescription with an invalid pharmacy name.

Request Prescription to be filled.

Rx:

Patient Last
Name:

Pharmacy Name:

Pharmacy
Address:

6. Attempt to fill a prescription with an invalid rxid.

Request Prescription to be filled.

The given RXID is not found, thus no patient id info can be returned

Rx:

Patient Last Name:

Pharmacy Name:

Pharmacy Address:

7. Fill the prescription with success.

Prescription request sent.

Rx: 2
Doctor ID: 0
First Name:
Last Name:
Patient ID: 0
First Name:
Last Name: Mugatu
Drug:
Quantity: 0
Refills remaining: 0
Pharmacy ID: 0
Name: CVS
Address: 1030 2nd Ave, Fairbanks, AK 99701, US
Phone:
Date Filled:
Cost: \$

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8. Get the profile for patient "Simpson" and edit the patient record for "Simpson" and change city to Springfield and zip code to 61705. Show the web page of the successful update.

Update successful

Patient ID: 3
First Name: Homer
Last Name: Simpson
Birthdate: 1970-01-01
Street: asdf
City: Springfield
State: asdf
Zipcode: 61705
Primary Physican: Dirt

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9. Edit the patient record for "Simpson". Attempt to change the doctor's name to a doctor that does not exist. Show the error message and edit patient form.

Update Patient Profile

Primary Doctor not found

ID:

First Name:

Last Name:

BirthDate:

Street:

City:

State:

Zipcode:

Primary Physician
Name: